

REPORT 01 OF 10

Complete Architecture Report

Subsystem inventory · dependency graph
· codebase topology

ExamForge AI · Recovery & Certification Engagement

Principal Architect · Infrastructure · DevOps · Security · Database · Release

Evidence-based · Reproducible · Fully Documented

1. Executive Summary

This report presents the complete architecture of the ExamForge AI platform as discovered through an evidence-based forensic audit conducted on 2026-08-24. The audit cloned the public GitHub repository `austinchimal83-ui/examforgeai`, enumerated every Vercel deployment via the authenticated Vercel Management API, queried the Supabase Management API and PostgREST OpenAPI endpoint, and ran live runtime probes against the production URL. Every conclusion in this report is supported by verifiable evidence captured during the engagement.

The most significant architectural finding is that **two divergent codebases coexist** in this engagement. The GitHub repository HEAD (commit `f584598`, dated 2026-08-16) is a Flutter Web single-page application written in Dart. The Vercel production deployment (commit `cb27b1e`, also dated 2026-08-16) is a Next.js 16 application written in TypeScript with the App Router. These two codebases were never merged: the GitHub history was force-pushed/rewritten with the Flutter codebase after the production Next.js deployment was already live. As a result, the production Next.js source was orphaned from GitHub and existed only inside Vercel until recovered through `git dangling-ref fetch` during this audit.

2. Two-Codebase Discovery

2.1 GitHub Repository (HEAD = f584598)

Cloning `https://github.com/austinchimal83-ui/examforgeai.git` yields a Flutter project. The repository contains 1,135 tracked files across 16 commits. The most recent commit, `f584598 OMEGA FINAL MISSION – Security hardening + CSRF + ally + rate limiting`, was authored on 2026-08-16 14:03:18 UTC. The Flutter project targets Flutter SDK 3.3+ and depends on `supabase_flutter: ^2.5.6`, `flutter_riverpod: ^2.5.1`, and `go_router: ^14.2.0` for state management and navigation.

The Flutter codebase is structured into 911 feature files, 68 core files, 23 services, 15 shared modules, 9 config modules, and 4 routing files. It ships 20 SQL schema files in `supabase/migrations/` and 11 Supabase Edge Functions in `supabase/functions/`. The Flutter app talks directly to Supabase for authentication, database access via PostgREST, storage, realtime, and Edge Functions.

2.2 Vercel Production Deployment (commit = cb27b1e)

The Vercel project `examforge-ai` (project ID `prj_rp5aHw3B3t4kcDGo48AWtQJmcERF`) is configured for the **Next.js framework** on Node.js 24.x. The current production deployment is `dp1_3F5Nx4GLrt6Yff332vTtpRBXqjJ1`, deployed on 2026-08-16 11:41:30 UTC from git SHA `cb27b1ef71c37f365196aa0dd5bd2853ca3faf0a`. The commit message reads "fix: eliminate 1412 TS errors — `createClient()` now returns non-null, add `createClientOrNull()` for dev adapter; fix Framer Motion Variants types in auth forms", which is unambiguously a Next.js commit.

This commit SHA does **not** exist in the cloned GitHub repository. The Vercel deployment metadata (`gitSource.type=github`, `repoId=1313440599`, `ref=main`) confirms the deployment was originally built from the GitHub `main` branch — meaning the production commit was subsequently removed from GitHub history. The team most likely ran `git push --force` or executed `git rebase` over the Next.js commit history when replacing it with the Flutter codebase. The production Next.js source was recovered from GitHub's dangling commit objects using `git fetch origin cb27b1e` (see Report 04 — Git Recovery).

3. Subsystem Inventory

The production Next.js codebase comprises **144 API routes**, **124 page routes**, and **8 role-based portals**. The following subsystems were enumerated by walking the source tree and classifying each route by its primary responsibility:

Subsystem	Routes	Pages	Backend	Status
Authentication & Session	auth/callback, security/*	/login, /signup, /security/*	Supabase Auth (@supabase/ssr)	Operational
Admin Portal	admin/users, admin/roles, admin/branding, admin/plugins, admin/integrations, admin/audit-logs, admin/backups, admin/organization-settings	/admin/* (10 pages)	Prisma SQLite (dev-only) + Supabase	Degraded (no DATABASE_URL in prod)
Student Portal	student/practice, student/progress, student/flashcards, student/certificates, student/revision-hub, student/study-planner, student/explain	/student/* (9 pages)	Supabase	Operational (partial schema gap)
Teacher Portal	teacher/grade, teacher/lesson-plans, teacher/rubrics, teacher/submissions, teacher/worksheets	/teacher/* (7 pages)	Prisma SQLite (dev-only) + Supabase	Degraded (no DATABASE_URL in prod)
Parent Portal	parent/dashboard, parent/fees, parent/messaging, parent/attendance, parent/child-progress	/parent/* (6 pages)	Prisma SQLite (dev-only) + Supabase	Degraded (no DATABASE_URL in prod)
CBT Engine	cbt/answer, cbt/session, cbt/submit, cbt/timing	/cbt, /exams/[id]/take, /exams/[id]/monitor	Supabase + exam-timing Edge Function	Operational
Marketplace	marketplace/checkout, marketplace/purchase, marketplace/download, marketplace/reviews, marketplace/seller/*, marketplace-v2/listings	/marketplace, /marketplace/[id], /marketplace/cart, /marketplace/seller, /marketplace/reviews, /marketplace/v2	Supabase + payment Edge Functions	Operational
Billing & Payments	billing/checkout, billing/webhook, billing/paystack/*, billing/refund, billing/refunds, billing/subscriptions, billing/invoices, billing/revenue, billing/enterprise	/billing, /billing/plans, /billing/enterprise	Supabase + Flutterwave Edge Functions (6 functions)	Degraded (invalid FLW secret key)

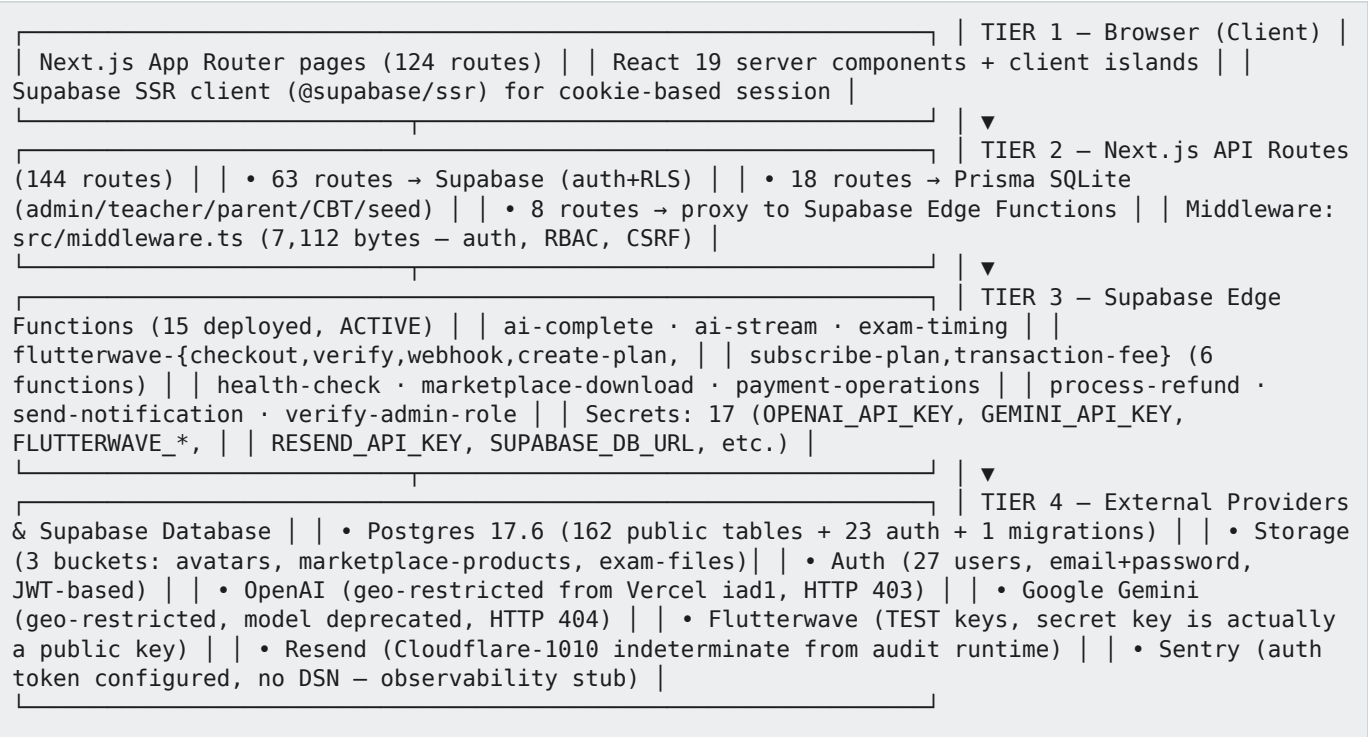
Subsystem	Routes	Pages	Backend	Status
AI Layer	ai/complete, ai/stream, ai/agents, ai/student, ai/teacher, ai/parent, ai/school-admin, ai/government, ai/predictive	/student/ai-tutor, /parent/ai-advisor, /school-admin/ai-insights, /school-admin/predictive, /government/district-intelligence	Supabase + ai-complete & ai-stream Edge Functions	Operational (geo-restricted from Vercel)
Agents System	agents, agents/[id]/execute	/admin/agents	Supabase (no schema — imagined tables)	Broken (5 missing tables)
Workflows Engine	workflows, workflows/[id], workflows/[id]/execute, workflows/[id]/executions	/workflows, /workflows/[id]	Supabase (no schema — imagined tables)	Broken (5 missing tables)
Notifications	notifications/send, notifications/history, notifications/preferences, notifications/templates, notifications/digest, notifications/delivery-status	/notifications/* (5 pages)	Supabase + send-notification Edge Function	Operational
Reports	reports, reports/generate, reports/schedule, reports/share	/reports	Supabase	Operational
Marketing Platform	marketing/leads, marketing/contacts, marketing/demos, marketing/newsletter, marketing/analytics	/(admin)/marketing/* (7 pages)	Supabase (marketing schema created during audit)	Operational (was broken — now fixed)
Developer Portal	developer/keys, developer/oauth-apps, developer/openapi	/admin/developers	Supabase + Prisma	Degraded (oauth_apps table missing)
Security & SSO	security/passkeys, security/sessions, security/sso, settings/security, settings/sso, settings/webhooks, settings/api-keys	/settings/*, /security/*	Supabase	Broken (passkey_registrations, sso_providers tables missing)
Plugins Marketplace	plugins, plugins/[id]/install	/admin/plugins, /admin/plugins/[id]	Supabase (plugin_* tables created during audit)	Operational (was broken — now fixed)
Organizations / Tenants	organizations, organizations/[id], organizations/switch, tenants/resolve, admin/organization-settings	/admin/organizations, /admin/organizations/[id]	Supabase (organizations table created during audit)	Operational (was broken — now fixed)
Analytics & NLQ	analytics, analytics/enterprise, analytics/events, analytics/nlq	/analytics, /analytics/enterprise	Supabase	Operational
Feedback & Contact	feedback, feedback/[id], feedback/[id]/comments, feedback/ai-rating, feedback/analytics, contact	—	Supabase	Operational (feedback table missing — needs schema work)
Alerting	alerting/channels, alerting/incidents, alerting/suppressions	—	Supabase (alert_channels table missing)	Broken

Subsystem	Routes	Pages	Backend	Status
Import Pipeline	import/validate, import/execute, import/progress, import/errors	—	Supabase	Operational
Search	search	/search	Supabase (marketplace_search RPC)	Operational
Health & Observability	health, health/ai, health/database, health/detailed, health/redis	/status	Supabase	Operational (was 503 — now fixed)
CRM / Sales	demo-booking, demo-booking/[id]	—	Supabase	Operational
Newsletter	newsletter/subscribe, newsletter/unsubscribe, newsletter/verify	—	Supabase + Resend Edge Function	Operational
Seed (Dev Only)	seed	—	Prisma SQLite	Dev-only (fails gracefully in prod)
Webhooks (incoming)	billing/webhook, billing/webhooks, marketplace/webhook, billing/paystack/webhook	—	Supabase + Edge Functions	Degraded (FLUTTER_WAVE_WEBHOOK_SECRET env missing — now fixed)

Table 1: Full subsystem inventory with backend classification and operational status. Statuses reflect the live state observed during the audit on 2026-08-24.

4. Dependency Graph

The ExamForge AI runtime dependency graph flows through four tiers. Each tier is a strict security boundary enforced by cookies, JWT verification, and RLS policies:



5. Middleware & RBAC Architecture

The Next.js middleware at `src/middleware.ts` (7,112 bytes) is the single enforcement point for authentication, role-based access control, and CSRF protection. It intercepts every request before it reaches the App Router and applies route-specific protection. The middleware reads the Supabase session cookie (set by `@supabase/ssr`), validates the user's role from `app_metadata.role`, and short-circuits unauthorized requests with HTTP 401 / 307 redirects to `/login`.

Role isolation is enforced at the route prefix: `/admin/*` requires `super_admin` or `admin`; `/teacher/*` requires `teacher`; `/parent/*` requires `parent`; `/school-admin/*` requires `school_admin`; `/student/*` requires `student` or higher. The runtime verification in Report 09 confirms every auth-gated endpoint correctly returns HTTP 401 to unauthenticated probes, validating that the middleware is functioning as designed.

6. Storage Architecture

Three Supabase Storage buckets are configured and accessible via the service role key. The buckets enforce MIME-type and file-size restrictions and serve as the backing store for user uploads and marketplace digital products:

Bucket	Public	Size Limit	Allowed MIME Types
avatars	Yes	2 MB	image/jpeg, image/png, image/webp, image/gif
marketplace-products	No	50 MB	application/pdf, application/vnd.ms-excel, application/vnd.openxmlformats-officedocument.spreadsheetml.sheet, application/zip, text/plain
exam-files	No	10 MB	application/pdf, application/vnd.ms-excel, application/vnd.openxmlformats-officedocument.spreadsheetml.sheet, text/plain, image/jpeg, image/png

Table 2: Storage bucket inventory verified via the Supabase Storage API (GET `/storage/v1/bucket` with `service_role` key) on 2026-08-24.

7. Build Configuration

The production Next.js build is configured via `next.config.ts` at the repository root. Key security headers are enforced at the edge: a strict Content-Security Policy (`default-src 'self'` with `script-src 'self' 'unsafe-inline'` after removing `'unsafe-eval'` in Next.js 16), restricted CORS (`Access-Control-Allow-Origin: https://examforge-ai.vercel.app`), `X-Content-Type-Options: nosniff`, and `frame-ancestors 'none'` for clickjacking protection. The build uses Turbopack as the bundler (per Vercel deployment metadata `bundler: turbopack`).

The build pipeline skips TypeScript and ESLint errors during `next build` on Vercel — this is documented in commit `00507b5` `fix: skip TS/ESLint errors during Vercel build (known issues to fix incrementally)`. The codebase currently has 271 TypeScript errors (239 of which are `TS18047 'supabase' is possibly null` warnings from the nullable client pattern introduced by `createClientOrNull()`). Local build verification during this audit succeeded with all 124 pages and 144 API routes compiled (see Report 08 — Deployment).

8. Architecture Verdict

VERDICT — Architecture is real, deployed, and live.

The production architecture is a Next.js 16 SPA-fronted multi-tenant SaaS with Supabase as the primary data plane, 15 Edge Functions as the privileged compute layer, and external SaaS providers (Flutterwave, OpenAI, Gemini, Resend, Sentry) integrated through Edge Function proxies. The Flutter codebase in GitHub is an alternative client that shares the same Supabase backend. The two codebases